Alejandra C. *Hernández*

Senior Researcher · Sensing and Perception · IoT Technologies and Cyber-Physical Systems · Ericsson Research, Stockholm

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Robotics and computer vision researcher in **semantic perception for mobile robots**, with a background in robotics research since 2015, industrial R&D and university teaching. Current focus: **domain adaptation and domain shift detection** for open-vocabulary segmentation with vision foundation models, deployed on embedded hardware. Initiated and co-formulated a Digital Futures research project with KTH; inventor on numerous patent families; PhD cum laude.

EXPERIENCE

2023 – present

Senior Researcher

Ericsson Research, Stockholm, Sweden · Sensing and Perception team, IoT Technologies and Cyber-Physical Systems

- **Initiated and co-formulated** Seamless indoor-outdoor mobile robot semantic perception and navigation with domain adaptation (SIMNAV), a Digital Futures Industrial Innovation Project with KTH granted in 2025, and lead the Ericsson Research contribution as industrial partner.
- Research on **training-free domain adaptation** and **multi-signal domain shift detection** for open-vocabulary semantic segmentation, evaluated on real robot data indoors and outdoors and deployed on embedded platforms.
- Inventor on **numerous patent families** since 2021, most recently three applications filed in 2026 on domain adaptation and domain shift detection.
- Mentor MSc thesis students; yearly guest lecturer in the Service Robots course at Lund University.

2021 – 2023

Experienced Researcher

Ericsson Research, Stockholm, Sweden

- Research on object detection, localisation and mapping, and perception offloading over mobile networks for connected devices and robots; published in ACM TECS and robotics journals.

2015 – 2021

Researcher and Teaching Assistant

RoboticsLab, Universidad Carlos III de Madrid (UC3M), Spain · Dept. of Systems Engineering and Automation

- PhD research on **scene understanding for indoor service robots**: semantic and topological navigation, object-based localisation, place recognition and object search.
- Technical team member in national research projects (HEROITEA, NAVEGASE-AUTOCOGNAV, Robotics Digital Innovation Hub) and in industrial collaborations with FCC, Indra, Cartonplast and DOMO.
- Taught Mobile Robots, Industrial Automation I & II and Automated Production Systems at BSc and MSc level for five years.

2008 – 2014

Systems engineering roles, Venezuela

FUNDELEC (Foundation for the Development of the Electrical Service), Mercantil Bank, CANTV

Coordination assistant on an unmanned aerial vehicle for transmission-line inspection and on simulation software for national grid planning; earlier systems analyst and risk-systems roles.

EDUCATION

2017 – 2021

PhD in Electrical Engineering, Electronics and Automation

Universidad Carlos III de Madrid · cum laude, international doctorate mention · advisor Ramón Barber

Thesis: Scene Understanding for Autonomous Robots Operating in Indoor Environments. Research visits: DLR Institute of Robotics and Mechatronics (2018), CIIRC, Czech Technical University in Prague (2019), AASS, Örebro University (2020).

2014 – 2016

MSc in Robotics and Automation
Universidad Carlos III de Madrid · thesis proposed for honours

2003 – 2008

BSc in Systems Engineering
UNEXPO Antonio José de Sucre, Venezuela · cum laude, first in class

RESEARCH FUNDING AND PROJECTS

2025 – present

SIMNAV · Digital Futures Industrial Innovation Project
Ericsson Research and KTH Royal Institute of Technology · PI Patric Jensfelt; with Olov Andersson

Initiated and co-formulated the proposal; lead the Ericsson contribution. Funds a postdoc and part of a research engineer. Fast, scalable adaptation of segmentation models for robots that operate mostly outdoors but must also work indoors.

2015 – 2021

National projects at UC3M
HEROITEA, multi-robot assistance of elderly people (Spanish Ministry of Economy, 2019–2021); Robotics Digital Innovation Hub phases III–IV (Comunidad de Madrid, 2016–2021); NAVEGASE-AUTOCOGNAV, semantic outdoor navigation (2015–2017). Technical team member.

SELECTED PUBLICATIONS

arXiv 2026 under review, ICRA 2027

When to Adapt: Multi-Signal Domain Shift Detection for Efficient Training-Free Adaptation in Open-Vocabulary Segmentation
M. Antonazzi, A. C. Hernández, Y. Cai, J. Araújo, O. Andersson, P. Jensfelt

ICRA 2026 workshop

Domain Shift-Aware Training-Free Adaptation for Open-Vocabulary Segmentation in Robotic Perception
M. Antonazzi, A. C. Hernández, Y. Cai, J. Araújo, O. Andersson, P. Jensfelt

arXiv 2026

Rethinking the Semantic Classification of Indoor Places by Mobile Robots
Ó. Martínez Mozos, A. C. Hernández, C. Gómez, R. Barber

ACM TECS 2025

SmartTBD: Smart Tracking for Resource-constrained Object Detection
S. Zhou, A. C. Hernández, C. Gómez, W. Yin, M. Björkman

ACM TECS 2024

OffloaD: Detection Failure-based Scheduler for Offloading Object Detection
C. Gómez, D. Patkhullaev, A. C. Hernández

JINT 2023

Localization Exploiting Semantic and Metric Information in Non-static Indoor Environments
C. Gómez, A. C. Hernández, R. Barber, C. Stachniss

RAS 2022

Exploiting the Confusions of Semantic Places to Improve Service Robotic Tasks in Indoor Environments
A. C. Hernández, C. Gómez, R. Barber, Ó. Martínez Mozos

ECMR 2021

Searching for Objects in Human Living Environments Based on Relevant Inferred and Mined Priors
A. C. Hernández, M. Durner, C. Gómez, I. Grixa, O. Teikmanis, Z.-C. Márton

RAS 2021

Change Detection Using Weighted Features for Image-Based Localization
E. Derner, C. Gómez, A. C. Hernández, R. Barber, R. Babuška

IROS 2020

Efficient Object Search Through Probability-Based Viewpoint Selection
A. C. Hernández, E. Derner, C. Gómez, R. Barber, R. Babuška

IEEE RA-L 2020

Object-Based Pose Graph for Dynamic Indoor Environments
C. Gómez, A. C. Hernández, E. Derner, R. Barber, R. Babuška

ICRA 2020

Hybrid Topological and 3D Dense Mapping through Autonomous Exploration for Large Indoor Environments
C. Gómez, M. Fehr, A. Millane, A. C. Hernández, J. Nieto, R. Barber, R. Siegwart

Sensors 2019

Topological Frontier-Based Exploration and Map-Building Using Semantic Information
C. Gómez, A. C. Hernández, R. Barber

ECMR 2019

Indoor Scene Recognition Based on Weighted Voting Schemes
A. C. Hernández, C. Gómez, E. Derner, R. Barber

IFAC IAV 2019	MiNERVA: Toposemantic Navigation Model Based on Visual Information for Indoor Environments A. C. Hernández, C. Gómez, R. Barber
Sensors 2016	Object Detection Applied to Indoor Environments for Mobile Robot Navigation A. C. Hernández, C. Gómez, J. Crespo, R. Barber

About 30 peer-reviewed publications in total, including a book chapter in Applications of Mobile Robots (IntechOpen, 2018). Full list on Google Scholar and ORCID.

PATENTS

Inventor on numerous patent families in robot perception, localisation and mapping (Ericsson, unless noted), plus applications filed in 2026 on domain adaptation and domain shift detection. A selection; the full list is on Google Patents.

- Object Detection US20260229015A1
- Spatially Modeling an Environment US20260203920A1
- Compensating for Lost Data Intended for Object Detection EP4758858A1
- Computing Localization Uncertainty for Devices Operating in Dynamic Environments US20260043886A1
- Mobile Device Operative to Encode and Stream Video WO2025087524A1
- Map-Based Simultaneous Localization and Mapping WO2025056159A1
- Determining Location of a Device within an Environment Comprising Planar Surfaces WO2025045330A1
- 3D Reconstruction EP4479937A1
- Updating a Map for Simultaneous Localization and Mapping AR130202A1
- Context-Aware Identification of Unknown Objects in Image Frames WO2024199634A1
- Object Detection Failure Measure WO2024156361A1
- Selective Localization and Mapping of a Partially Known Environment WO2023204740A1
- Robotic System for Maintenance Tasks on Lampposts and Conical Vertical Structures ES2759519, FCC Industrial with UC3M, 2020, co-inventor

TEACHING, SUPERVISION AND SERVICE

Guest lectures	Service Robots course, Lund University, yearly since 2021.
Teaching	Universidad Carlos III de Madrid, 2016–2021: Mobile Robots (MSc), Automated Production Systems (MSc), Industrial Automation I and II (BSc). Contributor to the Mobile Robots OpenCourseWare course and to a teaching-innovation project on small practical exercises.
Supervision	MSc thesis students at Ericsson Research since 2021, including Davron Patkhullaev (offloading object detection) and Shihang Zhou (resource-constrained tracking), both leading to ACM TECS papers.
Reviewing	Robotics and computer vision conferences and journals, including IEEE ICRA and IEEE/RSJ IROS.
Events	Supported the local organisation of IROS 2018 and IEEE-RAS Humanoids 2015.

AWARDS AND SCHOLARSHIPS

- PhD cum laude with international doctorate mention UC3M, 2021
- MSc thesis proposed for honours UC3M, 2016
- MECD collaboration scholarship, Dept. of Systems Engineering and Automation UC3M, 2014–2015
- BSc cum laude, first in class UNEXPO, 2008
- Scholarships for academic achievement Gran Mariscal de Ayacucho Foundation 2005–2008; UNEXPO 2004–2007; Ministry of Education 2002–2003

SKILLS AND LANGUAGES

Research	Semantic segmentation, open-vocabulary perception, vision foundation models, domain adaptation and test-time adaptation, domain shift detection, object detection and tracking, semantic mapping, localisation and navigation.
Engineering	Python, C/C++, PyTorch, ROS, OpenCV, PCL, Gazebo, Docker, Linux; deployment on embedded GPUs (NVIDIA Jetson); MATLAB/Simulink; industrial automation (PLC, Siemens TIA Portal, ABB RAPID).
Languages	Spanish (native), English (fluent), Swedish (intermediate), German (basic).